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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 5: Code Division Multiple Access (CDMA): Spreading and Despreading

Program Code:

```
import numpy as np

import matplotlib.pyplot as plt

# Input data

data = np.array([1, 0, 1, 1])

code = np.array([1, -1, 1, -1])

# Convert data into +1 and -1

data_signal = np.where(data == 1, 1, -1)

# Spreading

spread = np.repeat(data_signal, len(code)) * np.tile(code, len(data))

# Despreading

despread = []

for i in range(len(data)):

    result = np.sum(spread[i*4:(i+1)*4] * code)

    despread.append(1 if result > 0 else 0)

# Display output

print("Original Data:", data)

print("Spreading Code:", code)

print("Spread Signal:", spread)

print("Despread Data:", despread)

print("\nResult: CDMA spreading and despreading completed successfully.")

# Plot graph
```

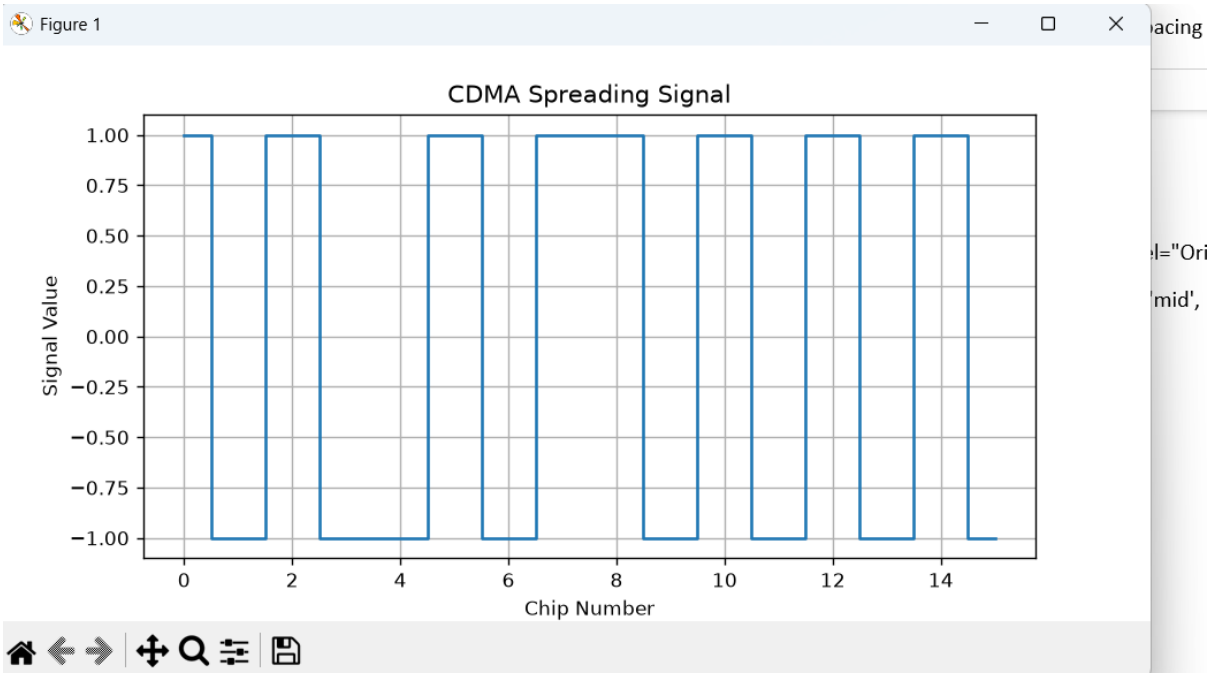
```
plt.figure(figsize=(8, 4))
plt.step(range(len(spread)), spread, where='mid')
plt.xlabel("Chip Number")
plt.ylabel("Signal Value")
plt.title("CDMA Spreading Signal")
plt.grid(True)
plt.show()
plt.figure(figsize=(8, 4))
plt.step(range(len(data)), data, where='mid', label="Original Data")
plt.step(range(len(despread)), despread, where='mid', label="Despread Data")
plt.xlabel("Bit Number")
plt.ylabel("Data")
plt.title("CDMA Despreading")
plt.legend()
plt.grid(True)
plt.show()
```

```
Hysteresis: 5 dB
Total Handoffs: 1
Handoff Time: [11]

Result: Handoff simulation completed successfully.

======== RESTART: D:/knowledge_Engineering/mobile computing/program0
Original Data: [1 0 1 1]
Spreading Code: [ 1 -1  1 -1]
Spread Signal: [ 1 -1  1 -1 -1  1 -1  1  1 -1  1 -1  1 -1  1 -1]
Despread Data: [1, 0, 1, 1]

Result: CDMA spreading and despreading completed successfully.
```


Figure 1
CDMA Spreading Signal
Signal Value
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10
12
14
Chip Number